

AlphaGenome + DeepSeek Agent Analysis

2026-06-03 · regvar v0.1.0

chr14_r.s1004030L None

Unranked



Assay	Gene	Biosample	Raw	Quantile
ChIP-seq TF	None	HEK293T	-1.36	-1.00
ChIP-seq TF	None	K562	-1.65	-1.00
ChIP-seq TF	None	HepG2	-1.46	-1.00
RNA-seq	MMP14	NCI-H460	-0.14	-1.00
RNA-seq	MMP14	HepG2	-0.16	-1.00
RNA-seq	MMP14	HeLa-S3	-0.15	-1.00
RNA-seq	MMP14	MCF-7	-0.13	-1.00
RNA-seq	MMP14	HT-29	-0.15	-1.00
RNA-seq	MMP14	GM23338	-0.16	-1.00
RNA-seq	MMP14	MCF-7	-0.13	-1.00
ChIP-seq TF	None	GM12878	-1.23	-1.00
RNA-seq	MMP14	HepG2	-0.16	-1.00
RNA-seq	MMP14	HepG2	-0.15	-1.00
RNA-seq	MMP14	Caco-2	-0.11	-1.00
RNA-seq	MMP14	K562	-0.14	-1.00

chr14:22836440:T>C (rs1004030) — ChIP-seq TF disruption | MMP14** This variant has the second-highest assay diversity. Strong ChIP-seq TF signal disruption in HEK293T, K562, HepG2, and GM12878 (abs quantile up to 0.99998), combined with **MMP14** (MT1-MMP) RNA-seq downregulation across NCI-H460, HepG2, HeLa-S3, MCF-7, HT-29. MMP14 is a membrane-tethered matrix metalloproteinase critical for extracellular matrix remodelling — directly relevant to stromal fibroblast invasion in the prostate TME. > **Predicted mechanism:** The C allele ablates a TF binding site

(predicted strong negative ChIP-seq TF scores), reducing MMP14 expression. Loss of MMP14 would impair fibroblast-mediated collagenolysis and matrix turnover, potentially promoting a desmoplastic stroma.